ECEN 662: Estimation & Detection Theory

Lecture 3: Exponential Family of Distributions

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Exponential family of distributions

- A model $f_\theta(x)$ is called an *exponential family of distributions* if it can be written as:

$$f_\theta(x) = h(x) \exp\left(\eta^t T - A(\eta)\right)$$

where:

  - $\eta = \eta(\theta)$ is an s-dimensional function of the model parameter θ
  - $T = T(x)$ is an s-dimensional function of the observation x
  - $h(x)$ is a one-dimensional function of the observation x known as the *base measure*
  - $A(\eta)$ is the *normalization* factor known as the *log-partition function* and can be computed as:

$$A(\eta) = \log\left(\int_{\mathcal{X}} h(x) \exp\left(\eta^t T(x)\right) dx\right)$$

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$$A(\eta) = \log\left(\int_{\mathcal{X}} h(x) \exp\left(\eta^t T(x)\right) dx\right)$$

- In addition, it is required that the *support set* $\{x : f_\theta(x) > 0\}$ does *not* depend on θ

Natural parameter and canonical representation

- Note that the likelihood function f_θ depends on the model parameter θ *only* through $\eta = \eta(\theta)$. Therefore, η is known as a *natural parameter* of the model, and any representation of the aforementioned form is known as a *canonical* representation of the exponential family

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- Note that for any given exponential family of distributions, canonical presentations are *not* unique. For example, if $(\eta, T, h, A(\eta))$ is a canonical representation, then

$$(\eta' = B\eta, T' = B^{-1}T, h' = h, A'(\eta') = A(B^{-1}\eta'))$$

  is also a canonical representation for any *invertible* matrix B

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- A canonical representation is called *irreducible* if neither $(\eta_i(\theta) : i \in [s])$ nor $(T_i(x) : i \in [s])$ satisfies a *linear-equality* constraint; otherwise, we can find an alternative canonical representation with a natural parameter of a *reduced* dimension

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- We say that an (irreducible) s-parameter exponential family is *full-ranked* if the space of the natural parameter $\eta(\Theta)$ contains an *s-dimensional* rectangle; otherwise, it is called a *curved* family

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- Let Θ be the space of the model parameter θ and let $\eta(\Theta) := \{\eta(\theta) : \theta \in \Theta\}$ be the space of the natural parameter η induced by $\eta = \eta(\theta)$
- We say that an (irreducible) s-parameter exponential family is *full-ranked* if the space of the natural parameter $\eta(\Theta)$ contains an *s-dimensional* rectangle; otherwise, it is called a *curved* family
- Note that by the Fisher-Neyman factorization theorem, $T(X)$ is a *sufficient statistic* for θ for any exponential family. In addition, if the exponential family is *full-ranked*, then $T(X)$ is also *complete* for θ

Multiple independent measurements

- Let $X = [X_1, \ldots, X_n]^t$, where

$$X_i \overset{iid}{\sim} f_\theta(x_i)$$

  and

$$f_\theta(x_i) = h(x_i) \exp\left(\eta^t T(x_i) - A(\eta)\right)$$

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$$f_\theta(x_i) = h(x_i) \exp\left(\eta^t T(x_i) - A(\eta)\right)$$

- We thus have

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i) = \left(\prod_{i=1}^{n} h(x_i)\right) \exp\left(\eta^t \sum_{i=1}^{n} T(x_i) - nA(\eta)\right),$$

which is (again) an exponential family of distribution with natural parameter η, sufficient statistic $\sum_{i=1}^{n} T(x_i)$, base measure $\prod_{i=1}^{n} h(x_i)$, and log-partition function $nA(\eta)$

Example: Bernoulli trials

- For Bernoulli trials, we have

$$
\begin{aligned}
p_\theta(x_i) &= \theta^{x_i}(1-\theta)^{1-x_i} \\
&= \exp\left(x_i \log\theta + (1-x_i)\log(1-\theta)\right) \\
&= \exp\left(x_i \log\frac{\theta}{1-\theta} - \log\frac{1}{1-\theta}\right)
\end{aligned}
$$

which is a one-parameter exponential family of distributions with

$$
\eta(\theta) = \log\frac{\theta}{1-\theta}, \quad T(x_i) = x_i, \quad h(x_i) = 1,
$$

$$
\text{and} \quad A(\eta) = \log\frac{1}{1-\theta} = \log\left(1+e^{\eta}\right)
$$

- Thus,

$$p_\theta(x) = \prod_{i=1}^{n} p_\theta(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\theta) = \log \frac{\theta}{1-\theta}, \quad T(x) = \sum_{i=1}^{n} x_i, \quad h(x) = 1,$$

$$\text{and} \quad A(\eta) = n \log(1 + e^{\eta})$$

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- Note that the space of the model parameter θ is $(0,1)$, and the space of the natural parameter

$$\eta = \log \frac{\theta}{1-\theta}$$

is $\mathcal{R}$

- Therefore, this exponential family is *full-ranked*, and the sufficient statistic

$$T(X) = \sum_{i=1}^{n} X_i$$

is also *complete*

Example: Gaussian with unknown mean

- For Gaussian with unknown mean, we have

$$f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x_i^2}{2\sigma^2}\right) \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right)$$

which is a one-parameter exponential family of distributions with

$$\eta(\mu) = \frac{\mu}{\sigma^2}, \quad T(x_i) = x_i, \quad h(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x_i^2}{2\sigma^2}\right)$$

$$\text{and} \quad A(\eta) = \frac{\mu^2}{2\sigma^2} = \frac{\sigma^2\eta^2}{2}$$

- Thus,

$$f_\mu(x) = \prod_{i=1}^{n} f_\mu(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\mu) = \frac{\mu}{\sigma^2}, \quad T(x) = \sum_{i=1}^{n} x_i,$$

$$h(x) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\sum_{i=1}^n x_i^2}{2\sigma^2}\right), \quad \text{and} \quad A(\eta) = \frac{n\sigma^2\eta^2}{2}$$

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- Note that the space of the model parameter μ is $\mathcal{R}$, and the space of the natural parameter

$$\eta = \frac{\mu}{\sigma^2}$$

is also $\mathcal{R}$

- Therefore, this exponential family is *full-ranked*. As a result, the sufficient statistic

$$T(X) = \sum_{i=1}^{n} X_i$$

is also *complete*

Example: Gaussian with unknown variance

- For Gaussian with unknown variance, we have

$$f_{\sigma^2}(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

which is a one-parameter exponential family of distributions with

$$\eta(\sigma^2) = -\frac{1}{2\sigma^2}, \quad T(x_i) = (x_i-\mu)^2, \quad h(x_i) = \frac{1}{\sqrt{\pi}}$$

$$\text{and} \quad A(\eta) = \frac{1}{2}\log(2\sigma^2) = -\frac{1}{2}\log(-\eta)$$

- Thus,

$$f_{\sigma^2}(x) = \prod_{i=1}^{n} f_{\sigma^2}(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\sigma^2) = -\frac{1}{2\sigma^2}, \quad T(x) = \sum_{i=1}^{n}(x_i - \mu)^2,$$

$$h(x) = \left(\frac{1}{\sqrt{\pi}}\right)^n, \quad \text{and} \quad A(\eta) = -\frac{n}{2}\log(-\eta)$$

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- Note that the space of the model parameter σ^2 is $\mathcal{R}_{++}$, and the space of the natural parameter

$$\eta = \frac{1}{2\sigma^2}$$

is $\mathcal{R}_{--}$

- Therefore, this exponential family is *full-ranked*. As a result, the sufficient statistic

$$T(X) = \sum_{i=1}^{n} (X_i - \mu)^2$$

is also *complete*

Example: Gaussian with unknown mean and variance

- For Gaussian with unknown mean and variance, we have

$$
\begin{aligned}
f_\theta(x_i) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{x_i^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2}\right) \\
&= \frac{1}{\sqrt{\pi}} \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{x_i^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\sigma^2)\right)
\end{aligned}
$$

which is a two-parameter exponential family of distributions with

$$
\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right), \quad T(x_i) = (x_i, x_i^2), \quad h(x_i) = \frac{1}{\sqrt{\pi}},
$$

and log-partition function

$$
A(\eta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2}\log(2\sigma^2) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-\eta_2)
$$

- Thus,

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i)$$

is also a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right),\ T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right),\ h(x) = \left(\frac{1}{\sqrt{\pi}}\right)^n,$$

and log-partition function

$$A(\eta) = -\frac{n\eta_1^2}{4\eta_2} - \frac{n}{2}\log(-\eta_2)$$

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$$A(\eta) = -\frac{n\eta_1^2}{4\eta_2} - \frac{n}{2}\log(-\eta_2)$$

- Note that the space of model parameter $\theta = (\mu, \sigma^2)$ is $\mathcal{R} \times \mathcal{R}_{++}$ and the space of the natural parameter $\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$ is $\mathcal{R} \times \mathcal{R}_{--}$

- Therefore, this exponential family is *full-ranked*, and the two-dimensional sufficient statistic

$$T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is also *complete*

Example: $X_i \sim \mathcal{N}(\theta, \theta^2)$ for some unknown $\theta > 0$

- In this case,

$$f_\theta(x_i) = \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x_i-\theta)^2}{2\theta^2}\right)$$
$$= \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{x_i^2}{2\theta^2} + \frac{x_i}{\theta} - \frac{1}{2}\right)$$

which is a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right), \quad T(x) = \left(x_i, x_i^2\right), \quad h(x_i) = \frac{1}{\sqrt{\pi}},$$

and log-partition function

$$A(\eta) = \frac{1}{2} + \frac{1}{2}\log(2\theta^2) = \frac{1}{2} - \frac{1}{2}\log(-\eta_2)$$

- Thus,

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i)$$

is also a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right),\ T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right),\ h(x) = \left(\frac{1}{\sqrt{2\pi}}\right)^n,$$

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$$A(\eta) = \frac{n}{2} - \frac{n}{2}\log(-\eta_2)$$

- Note that the space of model parameter θ is $\mathcal{R}_{++}$ and the space of the natural parameter $\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right)$ is a *one-dimensional curve* in $\mathcal{R}^2$. Therefore, this exponential family is *curved*

- As discussed in the previous lecture, in this case, the two-dimension sufficient statistic

$$T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is *not* complete for any $n \geq 1$

Uniform with unknown support

- Recall that for each $i \in [n]$, the likelihood function

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Uniform with unknown support

- Recall that for each $i \in [n]$, the likelihood function

$$f_\theta(x_i) = \frac{1}{\theta} 1_{\{x_i \le \theta\}}$$

- Because the support of $f_\theta(x_i)$ depends on θ, this is *not* an exponential family of distributions
- As we have shown previously, for this example, it is

$$T(X) = \max_{i \in [n]} X_i$$

rather than

$$T(X) = \sum_{i=1}^{n} X_i$$

that is a sufficient statistic for θ

Distribution and moments of $T(X)$

- Let $f_\theta(x)$ be an exponential family such that

  $$f_\theta(x) = h(x) \exp\left(\eta^t T - A(\eta)\right)$$

  The distribution of T is given by

  $$f_\theta(t) = k(t) \exp\left(\eta^t t - A(\eta)\right)$$

  for some new base measure $k(t)$, i.e, it is again an *exponential* family

Distribution and moments of $T(X)$

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for some new base measure $k(t)$, i.e, it is again an *exponential* family

- For example, when $X = (X_1, \ldots, X_n)$ where each X_i is Bernoulli with an unknown parameter θ, we can write

$$p_\theta(x) = \exp\left(\eta T(x) - A(\eta)\right),$$

where $\eta = \log \frac{\theta}{1-\theta}$, $h(x) = 1$, $T(x) = \sum_{i=1}^n x_i$, and $A(\eta) = n \log\left(1 + e^\eta\right)$

- The sufficient statistic $T(X) = \sum_{i=1}^{n} X_i$ is binomial whose pmf can be written as

$$
\begin{aligned}
p_\theta(t) &= \left(\begin{array}{c} n \\ t \end{array} \right) \theta^t (1-\theta)^{n-t} \\
&= \left(\begin{array}{c} n \\ t \end{array} \right) \exp\left(\left(\log \frac{\theta}{1-\theta} \right) t - n \log \frac{1}{1-\theta} \right) \\
&= k(t) \exp(\eta t - A(\eta)),
\end{aligned}
$$

where

$$
k(t) = \left(\begin{array}{c} n \\ t \end{array} \right)
$$

- The moment and cumulant generating functions of T can be computed from the log-partition function $A(\eta)$ as:

$$\begin{aligned} M_T(u;\theta) &:= \mathbb{E}_\eta \left[e^{u^t T} \right] \\ &= \int_{\mathcal{T}} k(t) \exp\big((u+\eta)^t t - A(\eta)\big)\, dt \\ &= \exp(A(u+\eta) - A(\eta)) \end{aligned}$$

and hence

$$K_T(u;\theta) = \log M_T(u;\theta) = A(u+\eta) - A(\eta)$$

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  and hence

$$K_T(u;\theta) = \log M_T(u;\theta) = A(u+\eta) - A(\eta)$$

- It follows immediately that

$$\mathbb{E}_\theta[T] = \nabla A(\eta) \quad \text{and} \quad \mathrm{Cov}_\theta[T] = \nabla^2 A(\eta)$$

- Going back to the example of Bernoulli trials, we have $\eta = \log \frac{\theta}{1-\theta}$, $T(x) = \sum_{i=1}^{n} x_i$, and $A(\eta) = n \log (1 + e^{\eta})$

- Going back to the example of Bernoulli trials, we have $\eta = \log \frac{\theta}{1-\theta}$, $T(x) = \sum_{i=1}^{n} x_i$, and $A(\eta) = n \log(1 + e^{\eta})$
- We thus have

$$\mathbb{E}_\theta[T] = \nabla A(\eta) = \frac{ne^{\eta}}{1 + e^{\eta}} = n\theta$$

and

$$\mathrm{Var}_\theta[T] = \nabla^2 A(\eta) = \frac{ne^{\eta}}{(1 + e^{\eta})^2} = n\theta(1 - \theta)$$

Next lecture

Uniformly Minimum-Variance Unbiased Estimation